INFOTACT SOLUTIONS & CO.

Infotact Technical Internship Program: Data Analytics Project Specifications and Evaluation Criteria

The following document establishes the technical requirements, analytical directives, and sprint-based data development roadmaps for three distinct Data Analytics (DA) projects. Designed for execution within a fast-paced technology environment in Bengaluru, Karnataka, these specifications serve as comprehensive blueprints for data analytics interns entering the infotact startup ecosystem. The documentation provides structured frameworks intended to guide the analysis of large datasets, the creation of predictive models, and the deployment of interactive dashboards across three high-growth industry niches: Retail and E-commerce, Travel and Hospitality, and Healthcare Technology.

The primary directive of this documentation is to standardize the data lifecycle process, ensuring that all deliverables align with enterprise-grade data architecture patterns, rigorous version control protocols, and robust data privacy standards. Consequently, data analytics interns must operate at a professional level, demonstrating proficiency in data wrangling, statistical problem-solving, and business intelligence (BI) reporting.

This document outlines the requisite tasks, technological paradigms, and evaluation criteria for successful internship completion. It is imperative to note that the assessment of these projects relies heavily on the transparency and consistency of the development lifecycle. **The evaluation will only be done if the intern is having all 4 weeks of github commits and contributions.**

Project 1: Omnichannel Retail Sales and Inventory Analytics Dashboard

Executive Problem Statement

In the rapidly digitizing retail landscape, offline businesses transitioning to digital models frequently encounter systemic bottlenecks due to fragmented data across multiple sales channels. As multi-location retailers scale operations, they struggle to unify data from brick-and-mortar Point of Sale (POS) systems and online storefronts. This lack of centralized data leads to an inability to track overall sales trends, forecast inventory requirements, and identify top-performing product categories.

The objective of this project is to develop an end-to-end Data Analytics pipeline and an interactive Business Intelligence dashboard that unifies physical and digital retail data. The platform must process raw sales transactions, identify seasonal revenue patterns, and provide actionable business intelligence to optimize inventory turnover and targeted marketing.

Business Objectives and Key Performance Indicators

To ensure the solution delivers tangible enterprise value, the analysis must be oriented around specific operational metrics. The primary objective is to uncover hidden sales trends, calculate essential revenue metrics like Total Sales and Average Revenue per Order, and identify the peak purchasing periods throughout the day and week.

Data integrity is paramount. The intern must eliminate discrepancies by thoroughly cleaning the raw transaction logs, handling missing values, and standardizing data formats. The success of the project will be quantified by the dashboard's ability to seamlessly filter data across different timeframes (e.g., YoY, MoM growth), locations, and product categories to enable data-driven decision-making for inventory managers.

User Personas and Operational Workflows

Persona	Primary Operational Needs	System Interaction and Workflow
Store Manager	Localized sales insights, daily performance tracking, and identifying fast-moving inventory items.	Uses the dashboard to monitor daily revenue, compare weekday vs. weekend sales performance, and track specific product category trends.
Inventory Planner	Macroscopic visibility of stock movement, seasonality forecasting, and identifying low-performing products to minimize dead stock.	Analyzes historical data to predict Q4 holiday spikes and adjusts supply chain procurement strategies to prevent stockouts.
Regional Sales Director	High-level revenue aggregations, geographic performance comparisons, and ROI analysis.	Reviews interactive dashboards to evaluate the financial health of multiple retail nodes and makes overarching pricing strategy decisions.

Minimum Viable Product Specifications

The foundational feature of this project is the Data Cleaning and Transformation pipeline. The intern must intake raw CSV files containing variables like order_id, product, quantity, and price, and transform them into a structured format suitable for querying.

The core output is the Interactive Power BI / Tableau Dashboard. This must include dynamic visualizations addressing key business questions: the overall sales trend over a 12-month period, revenue contributions categorized by product lines, order volume per city, and a time-series analysis detailing peak operational hours. Furthermore, the intern must deliver a final written report detailing actionable insights, such as recommending targeted Q4 promotions based on historical data spikes.

Architectural Directives and Technology Stack

Component	Technology	Architectural Rationale
Data Preparation	Microsoft Excel / Pandas (Python)	Excel provides an intuitive interface for initial exploration (VLOOKUP, Pivot Tables), while Python's Pandas library handles automated, large-scale data cleaning and manipulation.
Database & Querying	SQL (MySQL or PostgreSQL)	SQL is essential for aggregating millions of rows efficiently, utilizing advanced JOINs, subqueries, and window functions to extract precise business metrics.
Data Visualization	Power BI or Tableau	Industry-standard BI tools necessary for creating highly interactive, shareable dashboards with Drill-down capabilities and automated data refreshing.

Four-Week Engineering Roadmap

The project execution is strictly divided into four weekly sprints, demanding rigorous adherence to the delivery schedule.

Week 1: Data Collection, Environment Setup, and Initial Cleaning

The inaugural week demands the establishment of the foundational infrastructure. Interns must initiate the GitHub repository, configuring it to track their Jupyter Notebooks and SQL scripts. The intern will import the raw retail dataset into Python/Excel to begin the Data Cleaning phase. Tasks include identifying and handling NULL values, removing duplicate records, standardizing date/time formats, and validating data types across all columns.

Week 2: Relational Database Design and SQL Aggregations

The second week focuses on data querying and metric extraction. The intern will import the cleaned dataset into a SQL database and write complex queries to answer the core business questions. This includes using `GROUP BY`, `ORDER BY`, and aggregate functions (`SUM`, `AVG`) to calculate Total Revenue, Best-Selling Products by Volume, and Geographic Sales Distribution. All SQL scripts must be committed to GitHub with clear, semantic commit messages.

Week 3: Dashboard Architecture and Data Visualization

Week three transitions the focus toward the user-facing Business Intelligence dashboard. The intern will connect Power BI or Tableau directly to the SQL database or the cleaned flat files. The primary deliverable is the creation of interactive charts (Line charts for time-series trends, Bar charts for product performance, and Heatmaps for peak hour analysis). The intern must implement slicers/filters allowing users to drill down by specific months or city locations.

Week 4: Insight Generation, Reporting, and Final Submission

The final week is dedicated to interpreting the visualizations and documenting actionable business strategies. The intern must analyze the dashboard to extract key findings (e.g., "Tuesdays generate the highest revenue," or "Q4 accounts for 33% of annual sales"). A final project report must be written outlining these insights and providing strategic recommendations. The GitHub repository must be finalized with a comprehensive `README.md` containing dataset descriptions, setup instructions, and high-resolution screenshots of the dashboard.

Project 2: Travel, Tourism & Hospitality - Customer Retention and Dynamic Pricing Analysis

Executive Problem Statement

In the highly competitive travel and hospitality sector, agencies and hotel chains suffer from significant revenue leakage due to unoptimized pricing and unpredictable customer cancellations. Traditional booking systems lack the predictive intelligence to anticipate when a

user is likely to cancel a reservation or to adjust room rates dynamically based on fluctuating demand and seasonality.

The objective of this project is to analyze historical booking datasets to uncover the primary drivers of customer cancellations (churn) and to explore seasonal demand trends. By deploying Exploratory Data Analysis (EDA) and basic predictive modeling, the intern will provide the data foundation necessary for a dynamic pricing engine and targeted customer retention campaigns.

Business Objectives and Key Performance Indicators

The strategic vision for this analysis centers on optimizing Revenue Per Available Room (RevPAR) and reducing the overall cancellation rate. The data pipeline must process historical booking records to identify patterns correlated with high cancellation probabilities (e.g., lead time, deposit type, or specific customer segments).

From an analytics perspective, success will be measured by the comprehensiveness of the EDA and the accuracy of the insights generated. The intern must successfully segment the customer base, visualize the booking curve (lead time vs. booking volume), and provide a statistical breakdown of pricing elasticity across different times of the year and geographic markets.

User Personas and Operational Workflows

Persona	Primary Operational Needs	System Interaction and Workflow
Revenue Manager	Maximizing daily revenue, understanding seasonal demand curves, and adjusting room pricing dynamically.	Reviews the pricing trend dashboards to identify periods of high demand elasticity, adjusting future rates to prevent unsold inventory.
Marketing Director	Identifying high-risk customer segments and optimizing promotional ad spend.	Analyzes the churn demographic data to design targeted email retention campaigns offering specialized discounts to users likely to cancel.

Data Scientist	Clean, well-structured datasets and baseline statistical models for future AI implementation.	Utilizes the intern's cleaned datasets and initial EDA scripts as the foundation for deploying advanced machine learning models in production.

Minimum Viable Product Specifications

The foundational feature is the Exploratory Data Analysis (EDA) Notebook. Utilizing a dataset containing hotel or flight booking records, the intern must perform univariate and bivariate analysis to determine the relationship between the Average Daily Rate (ADR) and cancellation rates.

The project requires the development of a Customer Segmentation Model, grouping users based on their booking behavior (e.g., Corporate vs. Leisure travelers, early planners vs. last-minute bookers). Finally, a Seasonal Pricing Dashboard must be built to visualize how pricing fluctuates week-over-week, providing a visual narrative of historical demand peaks and troughs.

Architectural Directives and Technology Stack

Component	Technology	Architectural Rationale
Data Processing & EDA	Python (Pandas, NumPy)	Python provides the most robust ecosystem for handling complex statistical analysis and data wrangling within a reproducible Jupyter Notebook environment.
Statistical Visualization	Matplotlib, Seaborn	Essential Python libraries for generating complex statistical graphics, such as correlation matrices, box plots for outlier detection, and distribution curves.

Predictive Analytics	Scikit-Learn	Allows for the implementation of baseline machine learning algorithms (like Logistic Regression) to predict binary outcomes, such as whether a booking will be canceled.

Four-Week Engineering Roadmap

Week 1: Data Acquisition, Cleaning, and Feature Engineering

The project initiates with the establishment of the Git repository. The intern will ingest a complex hospitality dataset (e.g., hotel booking demand). The primary task is rigorous data cleaning: handling missing values in critical columns (like `agent` or `company` IDs), treating extreme outliers in the Average Daily Rate, and engineering new features (such as calculating the total duration of stay by combining weekend and weekday nights).

Week 2: Exploratory Data Analysis (EDA) and Statistical Testing

Week two centers on deep exploratory analysis. The intern will utilize Seaborn and Matplotlib to visualize the data distribution. Crucial tasks include mapping the correlation matrix to identify which variables most strongly impact cancellations. The intern must document their findings directly within the Jupyter Notebook using Markdown cells, explaining the statistical significance of their visualizations.

Week 3: Baseline Predictive Modeling (Churn Prediction)

The third week transitions into predictive analytics. The intern will utilize Scikit-Learn to build a baseline classification model (e.g., Logistic Regression or a Decision Tree) to predict the likelihood of a booking cancellation. The intern must split the data into training and testing sets, train the model, and evaluate its performance using appropriate metrics such as Accuracy, Precision, Recall, and the ROC-AUC curve.

Week 4: Final Presentation, Dashboards, and Documentation

The final week is dedicated to synthesizing the technical findings into a business-facing presentation. The intern will export the cleaned data into Tableau or Power BI to create a final, interactive presentation dashboard. The GitHub repository must be finalized, ensuring all Jupyter Notebooks execute flawlessly from top to bottom, and the `README.md` clearly states the business problem, methodologies used, and the final strategic recommendations to the Revenue Management team.

Project 3: Healthcare & Medical - Chronic Disease Prediction and EHR Analysis

Executive Problem Statement

The global healthcare sector generates massive amounts of data through Electronic Health Records (EHR), yet much of this data remains underutilized for proactive patient care. Hospitals struggle to identify at-risk patients early, leading to higher treatment costs and worse patient outcomes. Furthermore, analyzing sensitive medical data requires strict adherence to ethical guidelines and privacy standards (like HIPAA) to ensure patient confidentiality.

This project demands the end-to-end data analysis of an anonymized Electronic Health Records dataset to predict the likelihood of Chronic Kidney Disease (CKD) or heart disease. The intern will build a predictive analytics pipeline that cleans messy medical data, identifies crucial biomarker correlations, and assists healthcare professionals in early detection and intervention planning.

Business Objectives and Key Performance Indicators

The uncompromising priority for this application is Data Ethics and Privacy. The intern must ensure all analysis is performed on strictly de-identified, synthetic, or anonymized open-source data, demonstrating a clear understanding of healthcare compliance.

The operational objective is to develop a predictive model that accurately classifies patients as high or low risk for a specific chronic disease based on their medical attributes (e.g., blood pressure, glucose levels, serum creatinine). Success will be measured by the model's evaluation metrics—specifically achieving a high Recall rate, as false negatives in healthcare (failing to identify a sick patient) carry severe clinical consequences.

User Personas and Operational Workflows

Persona	Primary Operational Needs	System Interaction and Workflow
Clinical Practitioner	Early warning systems for at-risk patients and clear identification of contributing health factors.	Reviews the analytical reports to understand which biomarkers (e.g., elevated fasting blood sugar) most strongly correlate with disease onset.

Hospital Administrator	Population health metrics and resource allocation planning.	Utilizes aggregate dashboards to view disease prevalence across different demographic segments to allocate clinical resources efficiently.
Data Ethics Officer	Ensuring that all predictive models are unbiased and patient data remains secure and anonymized.	Audits the data pipeline and GitHub repository to ensure no Protected Health Information (PHI) is inadvertently exposed or used improperly.

Minimum Viable Product Specifications

The application foundation is the Medical Data Preprocessing Module. Healthcare datasets are notoriously noisy; the intern must handle missing numerical values (e.g., using mean/median imputation) and encode categorical medical variables appropriately.

The core is the Machine Learning Classification Model. The intern will test multiple algorithms (e.g., Logistic Regression, Random Forest Classifier) to predict the binary target variable (Disease Presence: Yes/No). Concluding the workflow is the Healthcare Insights Dashboard, a visual representation highlighting the most significant features (Feature Importance) contributing to the model's predictions, providing transparency and interpretability to medical staff.

Architectural Directives and Technology Stack

Component	Technology	Architectural Rationale
Data Cleaning & Analysis	Python (Pandas, Scikit-Learn)	Pandas handles the rigorous manipulation of messy EHR data, while Scikit-Learn provides a comprehensive suite of robust classification algorithms and evaluation metrics.

Visualization	Matplotlib, Seaborn, Tableau	Seaborn excels at complex statistical plotting (like violin plots for biomarker distributions), while Tableau offers top-tier executive dashboarding.
Version Control & Ethics	Git, GitHub, Jupyter	Jupyter Notebooks allow for narrative, step-by-step documentation of the analytical process, which is critical for medical auditing and peer review.

Four-Week Engineering Roadmap

The execution of this project requires extreme attention to detail regarding data handling and statistical validity.

Week 1: Ethical Data Sourcing, Preprocessing, and Exploration

The project begins with the establishment of a secure GitHub repository. The intern must source an appropriate, anonymized medical dataset (e.g., from the UCI Machine Learning Repository). The critical task for week one is data preprocessing: assessing missing values, converting data types, and performing initial summary statistics. The intern must document their approach to handling missing medical data, explaining the clinical rationale behind their imputation strategy.

Week 2: Biomarker Correlation and Exploratory Data Analysis

Week two focuses on discovering relationships within the data. The intern will utilize visualization libraries to perform Exploratory Data Analysis (EDA). This involves plotting the distributions of various biomarkers (e.g., age vs. blood pressure, cholesterol levels) across the healthy and diseased patient cohorts. The intern must generate a correlation heatmap to identify highly correlated features and avoid multicollinearity before feeding the data into a model.

Week 3: Predictive Modeling and Algorithm Selection

The third week introduces machine learning implementation. The intern will divide the dataset into training and testing splits. They will then build and train at least two different classification models (e.g., Logistic Regression and Random Forest). The intern must tune the hyperparameters of these models to optimize performance, paying special attention to handling any class imbalances (e.g., utilizing SMOTE if the number of diseased patients is significantly lower than healthy patients).

Week 4: Clinical Evaluation, Dashboarding, and Final Deployment

The final sprint is dedicated to model evaluation and clinical reporting. The intern must evaluate the models using a Confusion Matrix, Precision, Recall, and the ROC-AUC score, explicitly explaining why Recall is the most critical metric for this specific medical use case. The intern will then export the finalized results to Tableau or Power BI to create an executive dashboard summarizing the patient risk profiles. Finally, the GitHub repository must be updated with a comprehensive Readme detailing the ethical considerations, the workflow, and instructions to reproduce the findings.

Universal Data Analytics and Version Control Guidelines

To simulate a rigorous, enterprise-grade data environment indicative of top-tier product companies in Bengaluru, strict adherence to modern analytical practices and version control is an absolute requirement. The core tenet of this internship program is continuous, transparent, and high-quality contribution. Analysis is not merely the final dashboard; it is the iterative, documented process of problem-solving.

Mandatory Evaluation Protocol

The evaluation of the intern's performance and the final project submission is governed by a strict, non-negotiable directive regarding version control consistency.

The evaluation will only be done if the intern is having all 4 weeks of github commits and contributions.

A project submitted with a compressed commit history—for example, uploading the entire Jupyter Notebook and all datasets in a massive, monolithic push during the final week or on the final day—will result in immediate disqualification. Professional data science is a continuous process of iteration, and the version control history must accurately reflect daily data cleaning progress, EDA pivots, model tuning, and debugging efforts.

GitHub Operational Standards and Data Security Practices

To pass the technical evaluation, interns must internalize and strictly adhere to the following Git and GitHub operational standards.

1. Commit Frequency, Granularity, and Semantic Messaging

Commits act as the immutable ledger of an analyst's thought process. Interns are expected to commit their code and notebook updates frequently, breaking down complex data transformations into logical, highly isolated units of work. A minimum of three to five meaningful commits per active development day is the expected baseline.

Commit messages must communicate the exact intent and context of the analytical change. Interns must utilize structured prefixes such as `data-clean:`, `eda:`, `model:`, or `docs:` to maintain a

highly readable repository history (e.g., `model: added Random Forest classifier and output ROC-AUC metrics`).

2. Notebook Management and Branching Strategy

Direct commits to the `main` or `master` branch are discouraged for major feature additions. All new analysis (e.g., testing a new machine learning algorithm) should occur on an isolated feature branch. Furthermore, when committing Jupyter Notebooks (`.ipynb` files), interns should ensure that output cells containing massive amounts of raw data or sensitive visualizations are cleared if they bloat the repository size or violate data privacy standards.

3. Data Privacy and Credential Security

Security hygiene is paramount, especially when handling datasets mimicking financial or healthcare information. Interns must utilize `.gitignore` files to ensure that massive raw datasets (e.g., raw `.csv` or `.xlsx` files exceeding GitHub's recommended limits) are not pushed to the repository. Hardcoding database credentials, API keys, or cloud access tokens within Python scripts or notebooks is grounds for immediate failure. All sensitive connection strings must be injected via secure Environment Variables.

By strictly enforcing these comprehensive specifications, the program ensures that deliverables transcend mere academic exercises, transforming into robust, scalable, and highly professional data analytics portfolios indicative of elite industry standards.